Michael Psenka

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About Me

I'm a 4th year PhD student in EECS (BAIR) at UC Berkeley, advised by Prof. Aditi Krishnapriyan, and was a pure math undergraduate at Princeton (worked on Riemannian geometry and optimization). *I specialize in mathematical approaches to pioneering deep learning algorithms*, with a current focus on advancing flow-based models like diffusion models. My work spans **disentangled representation learning, computer vision, reinforcement learning, and molecular dynamics**. I aim to use mathematical insights to develop adaptive deep learning methods that can automatically adjust to new settings and scenarios–essential for building robust AI applications that operate across diverse conditions in the real world. Recognizing that data growth is plateauing, my goal is to maximize the utility of existing data through precise mathematical understanding of model functions, their features, and their interactions with data, thereby enhancing AI adaptability and robustness in practical applications.

Education

University of California, Berkeley *Berkeley, CA*
MS/PhD in Electrical Engineering and Computer Science *Sept 2021 - Current*

- *Coursework*: deep unsupervised learning, nonlinear systems and control, 3D vision, high-dimensional data analysis
- **GPA**: 4.0/4.0

Princeton University *Princeton, NJ*
BA in Mathematics, certificates in Applied Math and Computer Science *Sept 2017 - June 2021*

- *CS Coursework: (grad)*: machine learning, deep learning, weakly supervised learning, reinforcement learning, information theory, complexity theory
- *Math Coursework: (grad)*: stochastic calculus, geometric PDE, general relativity, quantum stat mech. *(undergrad)*: probability theory, real/complex analysis, representation theory
- **GPA**: 3.6/4.0

Publications & Workshop Proceedings

M. Psenka*, A. Escontrela*, P. Abbeel, and Y. Ma (2024). **Learning a Diffusion Model Policy from Rewards via Q-Score Matching**. *ICML 2024*. Link to paper.

M. Psenka, D. Pai, V. Raman, S. Sastry, and Y. Ma (2024). **Representation Learning through Manifold Flattening and Reconstruction**. *JMLR*. Link to paper.

N. Rahmanian, M. Gupta, R. Soatto, S. Nachuri, **M. Psenka**, Y. Ma, and S. Sastry (2023). **Role of Uncertainty in Anticipatory Trajectory Prediction for a Ping-Pong Playing Robot**. *arXiv*. Link to paper.

D. Pai, **M. Psenka**, C.-Y. Chiu, M. Wu, E. Dobriban, and Y. Ma (2023). **Pursuit of a discriminative representation for multiple subspaces via sequential games**. *Journal of the Franklin Institute*. Link to paper.

X. Dai, S. Tong, M. Li, Z. Wu, **M. Psenka**, K. H. R. Chan, P. Zhai, Y. Yu, X. Yuan, H.-Y. Shum, et al. (2022). **CTRL: Closed-Loop Transcription to an LDR via Minimaxing Rate Reduction**. *Entropy Journal*. Link to paper.

R. Arbon*, M. Mannan*, **M. Psenka***, and S. Ragavan* (2022). **A Proof of The Triangular Ashbaugh-Benguria-Payne-Pólya-Weinberger Inequality**. *Journal of Spectral Theory*. Link to paper.

M. Psenka and N. Boumal (2020). **Second-order optimization for tensors with fixed tensor-train rank**. *NeurIPS OPT 2020 Workshop*. Link to paper.

M. Psenka, T. Birdal, and L. Guibas (2020). **Reconstruction Without Registration**. *IROS 2020 geometric methods workshop*. Link to paper.

Awards

2020	**Peter A. Greenberg '77 Memorial Prize**, won for solving an open problem in spectral geometry. Awarded for outstanding accomplishments in Mathematics by juniors	*Princeton*
2018	**HackPrinceton First Place**, won first place at intercollegiate hackathon for developing A.I.D.A.N., a chatbot that lets users interact with their dataset with statistical and machine learning tools. Link to project.	*Princeton*
2018	**Manfred Pyka Memorial Prize**, awarded to outstanding Physics undergraduates who have shown excellence in course work and promise in independent research	*Princeton*
2021	**Sigma Xi Honors Society**, academic honors society for scientific research	*Princeton*

Work Experience

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Co-head instructor, lecturer *Berkeley, CA*
University of California, Berkeley *June 2022 - Aug 2022*
- Organized and taught lectures for CS 70, an undergraduate class for discrete math and probability theory
- **Link to class page**

Undergraduate researcher *Palo Alto, CA*
Stanford University *June 2020 - Aug 2020*
- Worked with Dr. Tolga Birdal on a novel approach to multi-view reconstruction in computer vision that bypasses pairwise view registration

Undergraduate researcher *Princeton, NJ*
Princeton University *June 2019 - Sept 2019*
- Worked with Prof. Nicolas Boumal funded by the National Science Foundation through award DMS-1719558
- Successfully developed a state-of-the-art method for computing analytic Hessians and second order optimization over tensor train manifolds

Software engineer *Charleston, SC*
Moovila, Inc. *June 2018 - Aug 2018, '17, '16*
- Developed a machine learning algorithm for workplace analytics, and improved search engine for quicker and more robust search
- Worked through a patent application, co-inventor in patent for proprietary software
- Worked closely with dev team, participating in stand-up and sprints regularly

Skills

Programming	Python (JAX, pytorch, numpy), C#, C, MATLAB, Java, HTML/CSS, JavaScript
Miscellaneous	Linux, Shell, LaTeX, Git, AWS

Interests

Piano	Played since I was 3. Grew up mostly classical, got into jazz playing at restaurants in middle/high school
Princeton Pianist Ensemble	All pieces arranged in-house, charity performances, virtual concerts during quarantine (link)
Snowboarding	The more trees, the better
Games	Smash bros, chess, or anything in between; always up to learn a new game